Year 7

REVISION PACK

Unit 3

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Unit 3 · Collecting like terms

Q	Working	Answer
simplifying expressions		
1.1	**(a)** $3 + 7 - 1 = 9$ **(b)** $-4 + 9 - 2 = 3$ **(c)** $6x - x = 5x$ $2y - 5y = -3y$ **(d)** $r^2 + 4r^2 = 5r^2$ $6r - 2r = 4r$	**(a)** $9x$ **(b)** $3p$ **(c)** $5x - 3y$ **(d)** $5r^2 + 4r$
1.2	**(a)** $5 + 2 - 3 = 4$ **(b)** $-2 + 8 - 3 = 3$ **(c)** $7x - 3x = 4x$ $5y - 2y = 3y$ **(d)** $m^2 + m^2 = 3m^2$ $3m - m = 2m$	**(a)** $4y$ **(b)** $3q$ **(c)** $4x + 3y$ **(d)** $3m^2 + 2m$
1.3	**(a)** $8 + 4 - 5 = 7$ **(b)** $-5 + 11 - 4 = 2$ **(c)** $9x - 4x = 5x$ $3y - y = 2y$ **(d)** $n^2 + 5n^2 = 6n^2$ $7n - 4n = 3n$	**(a)** $7a$ **(b)** $2b$ **(c)** $5x + 2y$ **(d)** $6n^2 + 3n$
1.4	**(a)** $2 + 9 - 6 = 5$ **(b)** $-3 + 10 - 6 = 1$ **(c)** $5x - 2x = 3x$ $6y - 4y = 2y$ **(d)** $t^2 + 2t^2 = 5t^2$ $2t - t = t$	**(a)** $5m$ **(b)** k **(c)** $3x + 2y$ **(d)** $5t^2 + t$

1.5	**(a)** $10 + 3 - 7 = 6$ **(b)** $-6 + 12 - 1 = 5$ **(c)** $8x - 6x = 2x$ $9y - 3y = 6y$ **(d)** $x^2 + 6x^2 = 7x^2$ $8x - 3x = 5x$	**(a)** $6c$ **(b)** $5d$ **(c)** $2x + 6y$ **(d)** $7x^2 + 5x$
1.6	**(a)** $6 + 6 - 4 = 8$ **(b)** $-1 + 7 - 2 = 4$ **(c)** $4x - 3x = x$ $7y - 6y = y$ **(d)** $y^2 + y^2 = 5y^2$ $5y - 2y = 3y$	**(a)** $8z$ **(b)** $4w$ **(c)** $x + y$ **(d)** $5y^2 + 3y$

Unit 3 · Writing expressions

Q	Working	Answer
writing expressions from words		
2.1	**(a)** **(b)** $2 \times n = 2n$ kg $2n \times 1000 = 2000n$ g **(c)** **(d)** $x + x + 2x + 3$	**(a)** $h + 4$ **cm** **(b)** $2n$ **kg,** $2000n$ **g** **(c)** $T = 3x + 2y$ **(d)** $4x + 3$ **cm**
2.2	**(a)** **(b)** $3 \times n = 3n$ kg $3n \times 1000 = 3000n$ g **(c)** **(d)** $x + x + 3x + 7$	**(a)** $h - 9$ **cm** **(b)** $3n$ **kg,** $3000n$ **g** **(c)** $T = 5x + 4y$ **(d)** $5x + 7$ **cm**
2.3	**(a)** **(b)** $4 \times n = 4n$ kg $4n \times 1000 = 4000n$ g **(c)** **(d)** $x + x + 4x + 2$	**(a)** $h + 12$ **cm** **(b)** $4n$ **kg,** $4000n$ **g** **(c)** $T = 2x + 6y$ **(d)** $6x + 2$ **cm**

2.4	**(a)** **(b)** $10 \times n = 10n$ kg $10n \times 1000 = 10000n$ g **(c)** **(d)** $x + x + 2x + 10$	**(a)** $h - 7$ **cm** **(b)** $10n$ **kg,** $10000n$ **g** **(c)** $T = 6x + 2y$ **(d)** $4x + 10$ **cm**
2.5	**(a)** **(b)** $6 \times n = 6n$ kg $6n \times 1000 = 6000n$ g **(c)** **(d)** $x + x + 5x + 1$	**(a)** $h + 15$ **cm** **(b)** $6n$ **kg,** $6000n$ **g** **(c)** $T = 4x + 5y$ **(d)** $7x + 1$ **cm**
2.6	**(a)** **(b)** $25 \times n = 25n$ kg $25n \times 1000 = 25000n$ g **(c)** **(d)** $x + x + 3x + 8$	**(a)** $h - 3$ **cm** **(b)** $25n$ **kg,** $25000n$ **g** **(c)** $T = 7x + 3y$ **(d)** $5x + 8$ **cm**

Unit 3 · Substitution

Q	Working	Answer
substituting into formulae		
3.1	**(a)** $8 \times 3 + 4 = 24 + 4 = 28$ **(b)** $-2 \times (-5) + (-5) = 10 - 5 = 5$ **(c)** $10 \times 40 = 400$ **(d)** $4^2 - 5 = 16 - 5 = 11$	**(a) 28** **(b) 5** **(c) 400** **(d) 11**
3.2	**(a)** $6 \times 4 + 7 = 24 + 7 = 31$ **(b)** $-4 \times (-3) + (-6) = 12 - 6 = 6$ **(c)** $15 \times 8 = 120$ **(d)** $6^2 - 9 = 36 - 9 = 27$	**(a) 31** **(b) 6** **(c) 120** **(d) 27**

3.3	**(a)** $3 \times 9 + 2 = 27 + 2 = 29$ **(b)** $-5 \times (-2) + (-3) = 10 - 3 = 7$ **(c)** $11 \times 7 = 77$ **(d)** $7^2 - 10 = 49 - 10 = 39$	**(a) 29** **(b) 7** **(c) 77** **(d) 39**
3.4	**(a)** $7 \times 5 + 6 = 35 + 6 = 41$ **(b)** $-3 \times (-6) + (-8) = 18 - 8 = 10$ **(c)** $9 \times 30 = 270$ **(d)** $3^2 - 4 = 9 - 4 = 5$	**(a) 41** **(b) 10** **(c) 270** **(d) 5**
3.5	**(a)** $4 \times 8 + 9 = 32 + 9 = 41$ **(b)** $-6 \times (-1) + (-4) = 6 - 4 = 2$ **(c)** $20 \times 12 = 240$ **(d)** $8^2 - 12 = 64 - 12 = 52$	**(a) 41** **(b) 2** **(c) 240** **(d) 52**
3.6	**(a)** $9 \times 2 + 5 = 18 + 5 = 23$ **(b)** $-2 \times (-7) + (-9) = 14 - 9 = 5$ **(c)** $14 \times 5 = 70$ **(d)** $10^2 - 1 = 100 - 1 = 99$	**(a) 23** **(b) 5** **(c) 70** **(d) 99**

Unit 3 · Substitution

Q	Working	Answer
substituting into fractions		
4.1	**(a)** $\dfrac{2 \times 3}{2} = \dfrac{6}{2} = 3$ **(b)** $\dfrac{7^2 - 1}{3} = \dfrac{48}{3} = 16$ **(c)** $\dfrac{4 \times 3 + 2}{7} = \dfrac{14}{7} = 2$	**(a) 3** **(b) 16** **(c) 2**

4.2	**(a)** $\frac{4\times5}{4}=\frac{20}{4}=5$ **(b)** $\frac{6^2-1}{5}=\frac{35}{5}=7$ **(c)** $\frac{4\times5+4}{6}=\frac{24}{6}=4$	**(a) 5** **(b) 7** **(c) 4**
4.3	**(a)** $\frac{2\times7}{2}=\frac{14}{2}=7$ **(b)** $\frac{5^2-4}{7}=\frac{21}{7}=3$ **(c)** $\frac{4\times7+2}{5}=\frac{30}{5}=6$	**(a) 7** **(b) 3** **(c) 6**
4.4	**(a)** $\frac{2\times4}{2}=\frac{8}{2}=4$ **(b)** $\frac{9^2-1}{4}=\frac{80}{4}=20$ **(c)** $\frac{4\times4+2}{9}=\frac{18}{9}=2$	**(a) 4** **(b) 20** **(c) 2**
4.5	**(a)** $\frac{3\times8}{3}=\frac{24}{3}=8$ **(b)** $\frac{5^2-1}{8}=\frac{24}{8}=3$ **(c)** $\frac{4\times8+3}{5}=\frac{35}{5}=7$	**(a) 8** **(b) 3** **(c) 7**
4.6	**(a)** $\frac{2\times9}{6}=\frac{18}{6}=3$ **(b)** $\frac{7^2-4}{9}=\frac{45}{9}=5$ **(c)** $\frac{4\times9+6}{7}=\frac{42}{7}=6$	**(a) 3** **(b) 5** **(c) 6**

Unit 3 · Expanding brackets

Q	Working	Answer
expanding a single bracket		

5.1	**(a)** $4 \times x + 4 \times 3$ **(b)** $x \times x + x \times 4$ **(c)** $3k \times 2k + 3k \times 4$ **(d)** $3 \times 2 \times (2 \times 2 + 4)$ $= 6 \times 8 = 48$	**(a)** $4x + 12$ **(b)** $x^2 + 4x$ **(c)** $6k^2 + 12k$ **(d) 48**
5.2	**(a)** $6 \times x + 6 \times 2$ **(b)** $x \times x + x \times 3$ **(c)** $4k \times 3k + 4k \times 1$ **(d)** $4 \times 2 \times (3 \times 2 + 1)$ $= 8 \times 7 = 56$	**(a)** $6x + 12$ **(b)** $x^2 + 3x$ **(c)** $12k^2 + 4k$ **(d) 56**
5.3	**(a)** $3 \times x + 3 \times 7$ **(b)** $x \times x + x \times 8$ **(c)** $5k \times 2k + 5k \times 3$ **(d)** $5 \times 1 \times (2 \times 1 + 3)$ $= 5 \times 5 = 25$	**(a)** $3x + 21$ **(b)** $x^2 + 8x$ **(c)** $10k^2 + 15k$ **(d) 25**
5.4	**(a)** $7 \times x + 7 \times 4$ **(b)** $x \times x + x \times 5$ **(c)** $2k \times 4k + 2k \times 7$ **(d)** $2 \times 3 \times (4 \times 3 + 7)$ $= 6 \times 19 = 114$	**(a)** $7x + 28$ **(b)** $x^2 + 5x$ **(c)** $8k^2 + 14k$ **(d) 114**
5.5	**(a)** $8 \times x + 8 \times 5$ **(b)** $x \times x + x \times 6$ **(c)** $6k \times 2k + 6k \times 1$ **(d)** $6 \times 2 \times (2 \times 2 + 1)$ $= 12 \times 5 = 60$	**(a)** $8x + 40$ **(b)** $x^2 + 6x$ **(c)** $12k^2 + 6k$ **(d) 60**

5.6	**(a)** $9 \times x + 9 \times 1$ **(b)** $x \times x + x \times 2$ **(c)** $3k \times 5k + 3k \times 2$ **(d)** $3 \times 2 \times (5 \times 2 + 2)$ $= 6 \times 12 = 72$	**(a)** $9x + 9$ **(b)** $x^2 + 2x$ **(c)** $15k^2 + 6k$ **(d) 72**

Unit 3 · Expressions for area

Q	Working	Answer
an expression for the area of a rectangle		
6.1	**(a)** **(b)** $A = y(y + 7)$ $= y^2 + 7y$ **(c)** $9^2 + 7 \times 9 = 81 + 63 = 144$	**(a)** $y + 7$ **cm** **(b) shown** **(c) 144 cm²**
6.2	**(a)** **(b)** $A = y(y + 3)$ $= y^2 + 3y$ **(c)** $8^2 + 3 \times 8 = 64 + 24 = 88$	**(a)** $y + 3$ **cm** **(b) shown** **(c) 88 cm²**
6.3	**(a)** **(b)** $A = y(y + 5)$ $= y^2 + 5y$ **(c)** $4^2 + 5 \times 4 = 16 + 20 = 36$	**(a)** $y + 5$ **cm** **(b) shown** **(c) 36 cm²**
6.4	**(a)** **(b)** $A = y(y - 2)$ $= y^2 - 2y$ **(c)** $10^2 - 2 \times 10 = 100 - 20 = 80$	**(a)** $y - 2$ **cm** **(b) shown** **(c) 80 cm²**
6.5	**(a)** **(b)** $A = y(y + 9)$ $= y^2 + 9y$ **(c)** $3^2 + 9 \times 3 = 9 + 27 = 36$	**(a)** $y + 9$ **cm** **(b) shown** **(c) 36 cm²**

6.6	**(a)** **(b)** $A = y(y-6)$ $= y^2 - 6y$ **(c)** $11^2 - 6 \times 11 = 121 - 66 = 55$	**(a)** $y - 6$ **cm** **(b) shown** **(c) 55 cm²**

Unit 3 · Perimeter expressions

Q	Working	Answer
perimeter expressions		
7.1	(a) $x + 2$ (b) $x + (x + 2) + x + (x + 2) = 4x + 4$ (c) $4 \times 7 + 4 = 32$	**(a)** $x + 2$ **(c) 32 cm**
7.2	(a) $x + 5$ (b) $x + (x + 5) + x + (x + 5) = 4x + 10$ (c) $4 \times 3 + 10 = 22$	**(a)** $x + 5$ **(c) 22 cm**
7.3	(a) $x + 1$ (b) $x + (x + 1) + x + (x + 1) = 4x + 2$ (c) $4 \times 9 + 2 = 38$	**(a)** $x + 1$ **(c) 38 cm**
7.4	(a) $x + 6$ (b) $x + (x + 6) + x + (x + 6) = 4x + 12$ (c) $4 \times 4 + 12 = 28$	**(a)** $x + 6$ **(c) 28 cm**
7.5	(a) $2x$ (b) $x + 2x + x + 2x = 6x$ (c) $6 \times 8 = 48$	**(a)** $2x$ **(c) 48 cm**
7.6	(a) $(x + 3) + (x + 3) + (x + 3) = 3x + 9$ (b) $3 \times 4 + 9 = 21$	**(b) 21 cm**